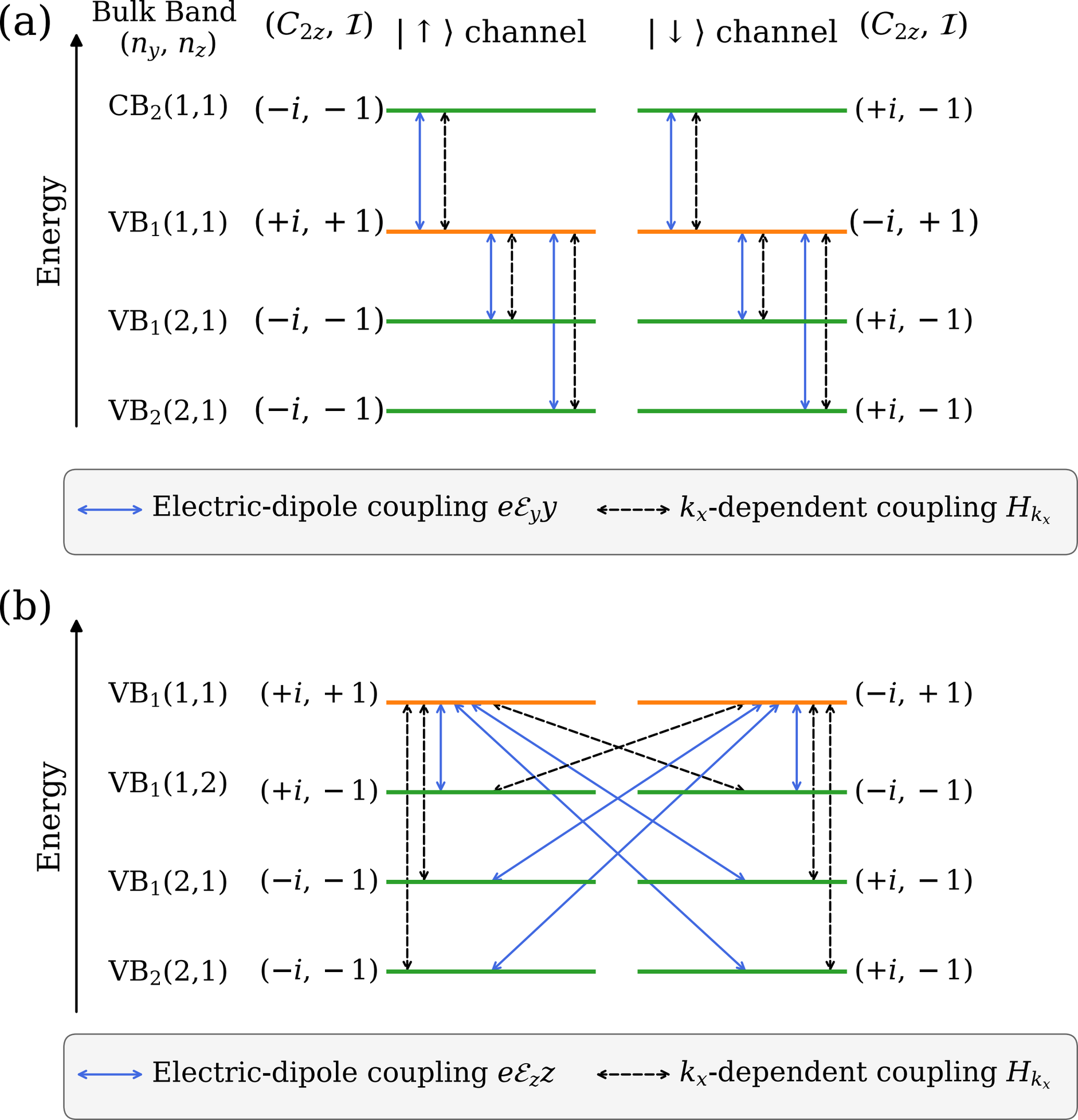

(a)
Bulk Band
(n_y, n_z)
$(C_{2z}, \mathcal{I})$
$|\uparrow\rangle$ channel
$|\downarrow\rangle$ channel
$(C_{2z}, \mathcal{I})$
Energy
$CB_2(1,1)$ $(-i, -1)$ $(+i, -1)$
$VB_1(1,1)$ $(+i, +1)$ $(-i, +1)$
$VB_1(2,1)$ $(-i, -1)$ $(+i, -1)$
$VB_2(2,1)$ $(-i, -1)$ $(+i, -1)$
Electric-dipole coupling $e\mathcal{E}_y y$
k_x-dependent coupling H_{k_x}
(b)
Energy
$VB_1(1,1)$ $(+i, +1)$ $(-i, +1)$
$VB_1(1,2)$ $(+i, -1)$ $(-i, -1)$
$VB_1(2,1)$ $(-i, -1)$ $(+i, -1)$
$VB_2(2,1)$ $(-i, -1)$ $(+i, -1)$
Electric-dipole coupling $e\mathcal{E}_z z$
k_x-dependent coupling H_{k_x}